

Integrating Quantum Computing with Multiconfiguration Pair-Density Functional Theory for Biological Electron Transfer

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Abstract

Accurate calculation of strongly correlated electronic systems requires proper treatment of both static and dynamic correlations, which remains challenging for conventional methods. To address this, we present VQE-PDFT, a quantum-classical hybrid framework that integrates variational quantum eigensolver with multiconfiguration pair-density functional theory (MC-PDFT). This framework strategically employs quantum circuits for multiconfigurational wavefunction representation while utilizing density functionals for correlation energy evaluation. The hybrid strategy maintains accurate treatment of static and dynamic correlations while reducing quantum resource requirements. Benchmark validation on the Charge-Transfer dataset confirmed that VQE-PDFT achieves results comparable to conventional MC-PDFT. Building upon this, we developed shallow-depth hardware-efficient ansatz circuits and integrated them into a QM/MM multiscale architecture to enable applications in complex biological systems. This extended framework, when applied to electron transfer in the European robin cryptochrome protein ErCRY4, yielded transfer rates that align well with experimental measurements. Importantly, successful execution on actual quantum hardware demonstrates practical feasibility for biological quantum computing applications, supported by comprehensive error analysis.

Keywords: Quantum Computing, Multiconfiguration, Electron Transfer, QM/MM

1 Introduction

Evaluating strongly correlated systems is one of the most fundamental challenges in modern science, where conventional theoretical approaches often struggle due to the intricate electronic coupling. The accurate description of these systems has led to major breakthroughs across multiple disciplines: in physics, it has driven the understanding of high-temperature superconductivity [1]; in chemistry, it has enabled the precise modeling of catalytic processes [2]; in biology, it has provided crucial insights into the mechanisms

of nitrogen fixation [3] and avian magnetoreception [4, 5].

From an electronic structure perspective, the challenge of evaluating strongly correlated systems lies in representing the multiconfigurational wavefunction [6]. Addressing this problem has led to the advent of several computational strategies, each with distinct advantages and limitations. Within the multiconfiguration self-consistent field (MCSCF) framework, complete active space configuration interaction (CASSI) [7] provides a foundation by performing configuration interaction within a predefined active space, though it lacks orbital optimization. Building upon this

foundation, the complete active space self-consistent-field (CASSCF) method enhances CASCI by simultaneously optimizing both the CI coefficients and orbital parameters, thereby more accurately capturing static correlation effects—the near-degeneracy effects arising from multiple electronic configurations of comparable energy [8, 9].

However, both CASCI and CASSCF often provide insufficient accuracy due to their incomplete treatment of dynamic correlation effects [8, 10]—the instantaneous electron-electron repulsion effects [11]. To remedy this deficiency, researchers have developed various post-SCF approaches that build upon MCSCF reference wavefunctions [12]. Complete active space second-order perturbation theory (CASPT2) [13, 14] represents one of the most successful examples, systematically recovering dynamic correlation, though it suffers from computational scaling limitations.

To address these computational limitations, multiconfiguration pair-density functional theory (MC-PDFT) [15, 16] offers a promising alternative by combining CASSCF-derived density matrices with an on-top density functional that capture dynamic correlation through opposite-spin electron pair probabilities at identical spatial coordinates. This hybrid approach greatly improves computational scaling relative to traditional post-SCF methods [6, 17].

However, MC-PDFT’s efficiency remains fundamentally constrained by its underlying CASSCF calculation. The exponential scaling of CASSCF with active space size renders calculations for large chemical systems computationally prohibitive, even with state-of-the-art hardware and advanced approximation schemes [18].

The emergence of quantum computing presents a potential solution to these exponential scaling challenges [19, 20]. Quantum algorithms naturally exploit superposition and entanglement to represent complex many-body wavefunctions, offering theoretical advantages for strongly correlated electronic systems [21]. Current noisy intermediate-scale quantum (NISQ) devices have spurred the development of hybrid quantum-classical algorithms [22], most notably the variational quantum eigensolver (VQE) [23] and its variants. These include ADAPT-VQE [24, 25], which dynamically constructs quantum circuits through iterative operator selection; Cluster-VQE [26], which reduces qubit requirements via system decomposition; and DMET-ESVQE [27], which achieves the combined advantages of both circuit optimization and qubit reduction through a distinct embedding-based approach.

Despite these advances, extending VQE-based methods to multiconfigurational systems remains problematic [28]. Existing approaches such as

UCCGSD, and k-UpCCGSD struggle with the simultaneous treatment of static and dynamic correlation [29], often requiring prohibitively deep quantum circuits that exceed current NISQ capabilities. Moreover, the inherent noise in NISQ devices raises fundamental questions about computational reliability [30, 31] for quantitative biological applications, necessitating error analysis in specific calculation. Therefore, the fundamental challenge lies in designing quantum algorithms that can efficiently capture multiconfigurational character while remaining compatible with near-term quantum hardware constraints.

This work introduces a novel quantum-classical hybrid framework that integrates VQE with multiconfiguration pair-density functional theory (MC-PDFT) for strongly correlated systems, referred to as VQE-PDFT. Our approach strategically employs quantum computation for multiconfigurational wavefunction representation while utilizing classical density functional methods for correlation energy evaluation, achieving reduced quantum resource requirements compared to existing multiconfigurational quantum algorithms [28, 29].

Initial validation on Charge-Transfer benchmark dataset confirmed that VQE-PDFT maintains the accuracy of conventional MC-PDFT while offering enhanced computational scalability. Building upon this foundation, we extended the framework to biological systems by developing specialized quantum circuits optimized for NISQ hardware constraints.

We demonstrated the practical utility of our methodology through comprehensive calculations of electron transfer processes in the European robin cryptochrome protein (ErCRY4). The computed transfer rates show excellent agreement with experimental ultrafast spectroscopy measurements [32], validating our quantum-classical approach for tackling strongly correlated systems in complex biological environments.

2 Results

2.1 The VQE integrated MC-PDFT

Our quantum-classical hybrid approach VQE-PDFT replaces the computationally expensive CASSCF optimization in MC-PDFT with VQE-based CASCI calculations. The modified workflow proceeds as follows: VQE optimizes a parameterized quantum circuit in CASCI framework to obtain a multiconfigurational ground state wavefunction, from which one-particle and two-particle reduced density matrices (1-RDM and 2-RDM) are extracted by measuring the corresponding matrix element expectation values on the quantum circuit as shown in Fig.1. These reduced density matrices are then utilized to compute the total energy following the MC-PDFT formalism.

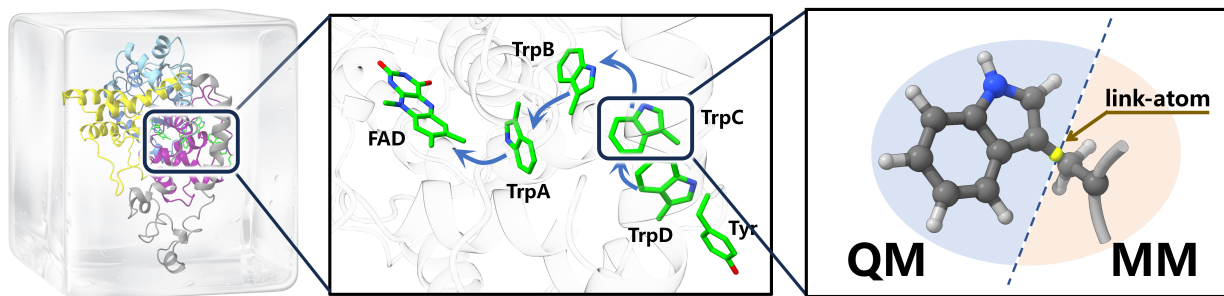


Fig. 2: Multiscale calculation framework of ErCRY4 protein. The quantum mechanical (QM) region is set to be two adjacent tryptophan residues, while the rest of the systems are considered in molecular mechanical (MM) level. The electrostatic embedding and link-atom methodology are adopted in evaluating interactions between the QM and MM regions.

We implemented electrostatic embedding and link-atom methodology to handle the interface between QM and MM regions, ensuring proper treatment of boundary effects [37]. This multiscale architecture extends the hybrid quantum computing pipeline previously developed by Li et al. [38], specifically adapting it for biological systems with complex electronic structure.

To demonstrate the practical utility of this multiscale framework, we selected the electron transfer process between adjacent tryptophan residues (TrpB and TrpC in Fig.2) in the European robin cryptochrome protein (ErCRY4) as our target application. This system presents an ideal test case due to its well-characterized experimental properties [5, 32] and the multiconfigurational nature. The electron transfer process involves photo-excitation of the FAD cofactor followed by sequential electron transfer among adjacent tryptophan residues, during which tryptophan residues alternate between neutral and cationic states. The cationic states exhibit open-shell electronic character with unpaired electrons, making the previously used standard UCCSD ansatz inadequate, as it assumes paired α and β electrons in each spatial orbital. This limitation, combined with NISQ device constraints, necessitates the development of specialized quantum circuits.

render it impractical for NISQ device implementation. To address this limitation, we developed shallow depth empirical hardware-efficient ansatz (HEA) circuits specifically optimized for biological electron transfer calculations.

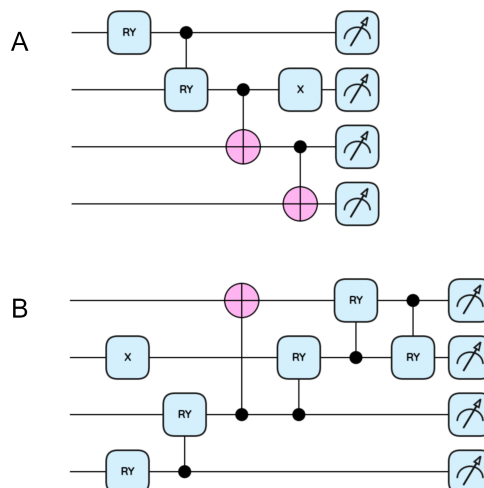


Fig. 3: The shallow-depth empirical ansatzes for electron transfer. **A** Closed-shell HEA circuit (CHEA) with circuit depth of 4. **B** Open-shell HEA circuit (OHEA) with circuit depth of 6. These two circuits are dramatically shallower than ROUCCSD with circuit depth of 45.

2.3 Shallow-depth empirical ansatz design

Conventional UCCSD ansatz performs well on evaluating closed-shell systems, and can be extended to restricted open-shell UCCSD (ROUCCSD) by explicitly considering single-electron occupations and the corresponding excitations to accommodate the unpaired electron scenarios in cationic tryptophan. However, ROUCCSD's prohibitively deep circuit structure and extensive non-local gate operations

Our approach employs distinct quantum circuits for the two electronic states encountered in the transfer process: a closed-shell HEA (CHEA, Fig.3A) for neutral tryptophan and an open-shell HEA (OHEA, Fig.3B) for cationic tryptophan. Both circuits were empirically designed to maintain the particle number conservation for each spin, while achieving minimal circuit depth compared to the ROUCCSD ansatz. The designing methodology is detailed in Section 4.2.

To validate the accuracy of the empirical HEA circuits, we randomly extracted a protein conformation from ErCRY4 molecular dynamics simulations reported in Ref. [5]. We then applied our multiscale framework by defining two adjacent tryptophan residues as separated QM regions for computational analysis. We employed a systematic approach that generates eight single-point energy calculations for two tryptophan residues, encompassing four open-shell cationic states and four closed-shell neutral states. These states are typically encountered in electron transfer processes, and the specific procedure of constructing these states follows the four-point scheme [39] (see also section 2.4.2).

For this validation study, we performed calculations using three different methods: the empirical HEA, ROUCCSD, and full configuration interaction (FCI), each applied to the active space within the CASCI framework. The comparative results are presented in Table 2.

	E_{FCI}	$E_{ROUCCSD}$	E_{HEA}	$\Delta E_{HEA-FCI}$
1	-2.1968	-2.1968	-2.1947	0.0021
2	-2.5535	-2.5535	-2.5500	0.0035
3	-2.1902	-2.1902	-2.1899	0.0004
4	-2.5424	-2.5424	-2.5391	0.0033
5	-2.1912	-2.1912	-2.1899	0.0013
6	-2.5621	-2.5621	-2.5584	0.0036
7	-2.1963	-2.1963	-2.1958	0.0005
8	-2.5310	-2.5310	-2.5279	0.0032

Table 2: Energy calculations on the activate space of tryptophan. The 1st, 3rd, 5th, and 7th single-point calculations are of open-shell with (3e,3o) activate space, while the others are of closed-shell with (4e,3o) activate space. The 6-31G basis set was used in all calculations. All energies are given in Hartree.

The results demonstrate that the proposed empirical HEA circuits achieve high accuracy with energy differences typically ranging from 10^{-4} to 10^{-3} Hartree with respect to the FCI’s results. Moreover, ROUCCSD energies match FCI values within excellent precision (at least $< 10^{-4}$ Hartree). This validation confirms that our HEA circuits maintain sufficient expressibility for accurate biological electron transfer calculations while offering substantial quantum resource savings.

2.4 Evaluate electron transfer in the ErCRY4

2.4.1 Marcus theory

Having demonstrated the accuracy of the empirical HEA circuits, we applied them to compute electron

transfer rates in the ErCRY4 protein system. Electron transfer kinetics in biological systems may be depicted by the Marcus theory, resulting the transfer rate being expressed as:

$$k_{ET} = \frac{2\pi}{\hbar} \frac{\langle |H_{DA}|^2 \rangle}{\sqrt{4\pi\lambda k_B T}} \exp \left[-\frac{(\Delta G^o + \lambda)^2}{4\lambda k_B T} \right], \quad (2)$$

which incorporates three key parameters: $\langle |H_{DA}| \rangle$, λ , and ΔG^o . The electronic coupling $\langle |H_{DA}| \rangle$ represents the interaction between two tryptophan residues, reflecting the orbital overlap and electronic communication pathways. Then the reorganization energy λ quantifies the energetic cost of nuclear rearrangement accompanying electron transfer. Lastly, the driving force ΔG^o corresponds to the thermodynamic bias for the transfer process, determined by the energy difference between reactant and product states.

Our VQE-PDFT framework within the multiscale QM/MM architecture enables direct evaluation of the reorganization energy and driving force.

2.4.2 Four-point scheme for λ and ΔG^o

With the Marcus theory framework outlined above, we now detail our computational approach. We focused specifically on the electron transfer between tryptophan residues TrpB (W372) and TrpC (W318) in the ErCRY4 protein. The initial state corresponded to cationic TrpB and neutral TrpC, while the final state involved neutral TrpB and cationic TrpC following electron transfer.

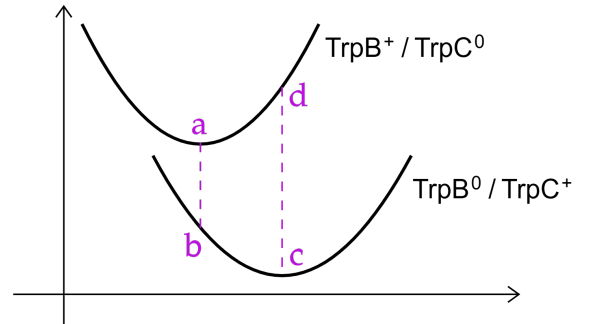


Fig. 4: The four-point scheme for evaluating reorganization energy and driving force.

a: $\text{TrpB}^+/\text{TrpC}^0$ state on the initial geometry; b: $\text{TrpB}^0/\text{TrpC}^+$ state on the initial geometry; c: $\text{TrpB}^0/\text{TrpC}^+$ state on the final geometry; d: $\text{TrpB}^+/\text{TrpC}^0$ state on the final geometry.

To quantify both the reorganization energy λ and driving force ΔG^o , we employed the four-point scheme [39], a well-established thermodynamic approach that captures the essential physics of Marcus theory. This method is based on the fundamental

assumption that the system responds instantaneously to changes in electronic charge distribution [40]. This approximation, although neglects potential non-Markovian effects arising from slower environmental responses [41], enables us to bypass costly dynamical simulations and focus directly on energetic differences between key electronic configurations.

As illustrated in Fig.4, the four-point scheme evaluates single-point energies at four distinct state-geometry combinations: each electronic state (initial and final) calculated at both its own optimized nuclear conformation and at the geometry optimized for the other state. This approach captures the energetic penalty arising from the mismatch between optimal nuclear arrangements for different charge distributions. Thereafter, the reorganization energy and the driving force of a given protein conformation might be extracted through:

$$\Delta G^o = |E_i^i - E_f^f|, \quad (3)$$

$$\lambda = |E_f^i - E_i^i| + |E_i^f - E_f^f|, \quad (4)$$

where $E_{state}^{geometry}$ denotes the energy of a given electronic state calculated at a specific nuclear geometry, while “ i ” and “ f ” stands for “initial” and “final” respectively.

To capture the dynamic behavior of the protein in biological environments, we sampled 20 distinct protein conformations from molecular dynamics simulations of ErCRY4 and applied the four-point scheme to each conformation using our VQE-PDFT/MM framework. We calculated λ and ΔG^o using both ROUCCSD and the proposed empirical HEA circuits, obtaining averaged value of $\lambda = 0.5701$ eV and $\Delta G^o = 0.0689$ eV for ROUCCSD, and $\lambda = 0.4356$ eV and $\Delta G^o = 0.0724$ eV for HEA, as listed in Table 3 and detailed in Supplementary Table S2 and S3. The good agreement between HEA and ROUCCSD results, considering the great reduction in circuit depth in HEA, validates the accuracy of the empirical HEA for biological applications while possessing substantial quantum resource savings.

2.4.3 Electronic coupling

With the reorganization energy and driving force determined, we now evaluate the electronic coupling $\langle |H_{DA}| \rangle$.

We computed the $\langle |H_{DA}| \rangle$ through transfer integral calculations between the TrpB and TrpC indole rings. The approach employs Boys localization [42] to transform canonical molecular orbitals into spatially localized orbitals centered on individual tryptophan sites, enabling clear identification of the relevant frontier orbitals for each fragment. This localization procedure is essential for defining well-separated

Ansatz	λ	ΔG^o	$\langle H_{DA} \rangle$	$\langle H_{DA} ^2 \rangle$
ROUCCSD	0.5701	0.0689	6.352E-3	1.1431E-4
HEA	0.4356	0.0724		

Table 3: The averaged λ , ΔG^o , and $\langle |H_{DA}| \rangle$. The reorganization energy and driving force are evaluated by four-point scheme, utilizing ROUCCSD as well as the empirical HEA, while the electronic coupling are determined by direct coupling scheme. The detailed results for individual configurations are provided in the Supporting Information. λ , ΔG^o , and $\langle |H_{DA}| \rangle$ are given in eV, while $\langle |H_{DA}|^2 \rangle$ is in (eV)². The 6-31G basis set and the tPBE functional were used in all calculations.

donor and acceptor orbitals in weakly coupled systems where standard canonical orbitals are typically delocalized across multiple sites.

The electronic coupling is extracted using the direct coupling (DC) scheme [43], which accounts for orbital non-orthogonality through:

$$H_{DA} = [T_{DA} - (e_D + e_A)S_{DA}/2] / (1 - S_{DA}^2), \quad (5)$$

where T_{DA} is the Hamiltonian coupling element, S_{DA} is overlap integral, and e_D, e_A are localized orbital energies for donor and acceptor sites, respectively.

Applying this methodology to the same 20 protein configurations, we obtained an averaged electronic coupling of $\langle |H_{DA}| \rangle = 6.35 \times 10^{-3}$ eV (6.35 meV) that aligns well with the 5 meV reported by Timmer et al. [32], also listed in Table 3 and detailed in Supplementary Table S4.

2.4.4 Electron transfer rate

With all Marcus theory parameters determined from the 20 sampled protein configurations, we proceeded to evaluate the electron transfer rates using Equation (2).

The computed rate were derived by substituting the averaged reorganization energies, driving forces, and electronic couplings into the Marcus expression (2). As a result, our empirical HEA approach yielded $k_{ET} = 0.944 \times 10^{10} s^{-1}$, aligning well with both ROUCCSD calculations ($0.864 \times 10^{10} s^{-1}$) and the experimental ultrafast transient absorption measurements [32] ($0.709 \times 10^{10} s^{-1}$), as shown in Figure 5. Notably, these results are approximately one order of magnitude slower than previous DFT predictions [5] ($5.0 \pm 1.8 \times 10^{10} s^{-1}$), as also pointed out in Ref. [32].

Experimental validation on quantum hardware

Moreover, to validate our approach on actual NISQ hardware, we executed the empirical HEA circuits on a superconducting quantum device for a randomly

selected protein conformation. With VQE’s optimization performed on a noiseless simulator and density matrix evaluations on quantum hardware with read-out error mitigation [44, 31], we obtained the single-point energies for the four-point scheme, eventually leading to the transfer rate $k_{ET} \approx 2.62 \times 10^8 s^{-1}$, as listed in Table 4.

Index	System	Empirical HEA on quantum hardware
1	point-a TrpB ⁺	-362.92966
2	point-a TrpC ⁰	-363.17866
3	point-b TrpB ⁰	-363.20554
4	point-b TrpC ⁺	-362.91475
5	point-c TrpB ⁰	-363.19954
6	point-c TrpC ⁺	-362.91126
7	point-d TrpB ⁺	-362.92451
8	point-d TrpC ⁰	-363.18130
λ		0.47021 eV
ΔG°		0.06764 eV
k_{ET}		$2.61742 \times 10^8 / s$

Table 4: Single-point energies, Marcus parameters, and the transfer rate of the single conformation. The conformation was randomly selected as the 2071 frame from the MD simulation and calculated in the QM/MM framework, with QM energies were extracted and listed here. The calculations were conducted by empirical HEA on superconducting quantum hardware. The single-point energies are given in Hartree. The 6-31G basis set and the tPBE functional were used in all calculations. Explicit single-point calculations and Marcus parameters of ROUCCSD and noiseless HEA could be found in Supplementary Information.

This result compares favorably with noiseless calculations: $1.89 \times 10^8 s^{-1}$ (HEA) and $2.49 \times 10^8 s^{-1}$ (ROUCCSD), demonstrating that quantum hardware may conduct electron transfer evaluations despite inherent noise. More explicit single-point energies and derived parameters are detailed in the Supplementary Table S5.

The successful hardware validation establishes the practical feasibility of our quantum computing framework for biological electron transfer calculations, representing a significant step toward real-world quantum computing applications in biochemistry.

2.5 Error analysis

The successful quantum hardware validation demonstrates practical feasibility, but might raise questions about the reliability of noisy quantum devices for the electron transfer process. To address these concerns, we conducted comprehensive error analysis comparing single-point energies and derived Marcus parameters across different computational approaches.

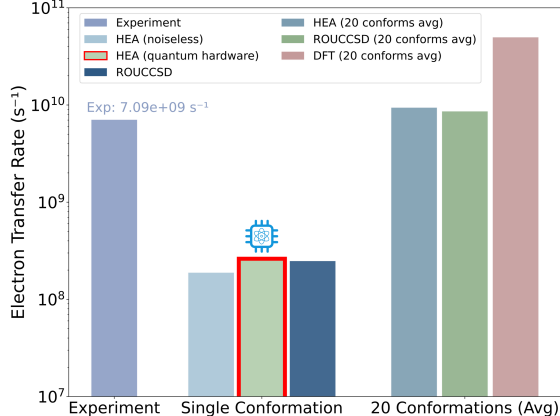


Fig. 5: Electron transfer rate comparison. The red marked transfer rate was evaluated on a quantum hardware. The single conformation results show rates one order of magnitude lower than averaged values of 20 conformations, reflecting typical conformational fluctuations in MD simulations where tryptophan positioning variations can alter transfer rates by several orders of magnitude [41]. The DFT prediction was obtained from Ref. [5]. Explicit parameters calculated for each conformations are listed in Supplementary Table S2.

2.5.1 Systematic error in single-point energies

To assess the computational accuracy of quantum hardware implementation, we examined single-point energy errors of the protein conformation (frame 2071) presented in Table 4. Our analysis employed ROUCCSD calculations as the reference standard, building upon our earlier validation that demonstrated excellent agreement between ROUCCSD and FCI results (Table 2). The quantum hardware calculations reveal systematic deviations from this ROUCCSD reference, as illustrated in Figure 6A and 6B, where all eight energy calculations exhibit consistent positive shifts.

This systematic bias indicates a non-random error source. The errors likely originate from correlated uncertainties in noisy matrix element measurements or from VQE-PDFT’s systematic response to noisy RDMs, rather than from purely random quantum fluctuations.

Moreover, the systematic nature of these hardware errors is noteworthy for biological electron transfer applications. Unlike random noise that would propagate unpredictably and through energy difference calculations, systematic errors might enable partial cancellation when computing the energy differences that underlie Marcus theory parameters. This observation motivates examining how these correlated single-point energy errors influence the

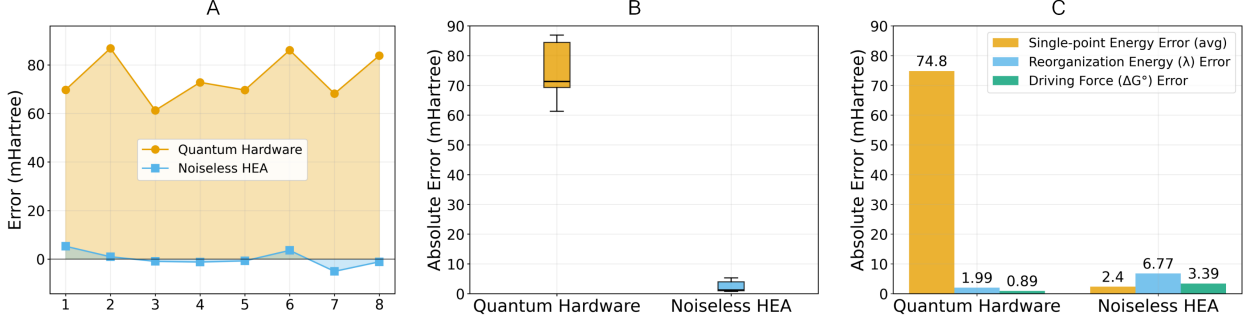


Fig. 6: Error analysis of single-point energies and Marcus parameters. ROUCCSD’s results are set as the comparing baseline. **A** Error comparison on 8 single-point energies of quantum hardware and noiseless empirical HEA, with quantum hardware’s results showing systematic errors (65.3 ± 14.2 mHartree positive deviation) compared to noiseless HEA relative to ROUCCSD reference. The system indices follow the index column in Table 4. **B** Error distributions of quantum hardware and noiseless empirical HEA. **C** Error Magnitude comparison between single-point energies and Marcus parameters. Explicit single-point calculation can be found in Supplementary Table S5.

derived reorganization energy and driving force that determine electron transfer rates.

2.5.2 Error cancellation in Marcus parameters

The feasibility of Marcus parameter calculations on quantum hardware can be understood through their mathematical structure. Both the driving force and reorganization energy depend on energy differences, as defined in Equations (3) and (4). Consider the driving force calculation on quantum hardware:

$$\begin{aligned} \Delta G^{o'} &= \left| E_i^{i'} - E_f^{f'} \right|, \\ &= \left| (E_i^i + \delta E_i) - (E_f^f + \delta E_f) \right|, \\ &= \left| (E_i^i - E_f^f) + (\delta E_i - \delta E_f) \right|, \end{aligned} \quad (6)$$

where systematic errors δE relative to the ROUCCSD reference partially cancel in the energy difference calculation, reducing the overall uncertainty to the net difference ($\delta E_i - \delta E_f$). A similar partial cancellation also functions in the reorganization energy calculation.

This error cancellation effect is demonstrated in our calculations, as illustrated in Figure 6B and 6C. While individual single-point energies exhibit errors of 61-86 mHartree on quantum hardware, the derived Marcus parameters show notable reduced absolute uncertainties: λ differs from the ROUCCSD reference by 2.0 mHartree, and ΔG^o by 0.9 mHartree. However, it should be noted that the relative errors of Marcus parameters actually increase due to their smaller absolute magnitudes ($\sim 12\%$ for λ , $\sim 36\%$ for ΔG^o).

Despite these relative error increases, the systematic nature of quantum hardware errors enables partial cancellation when computing energy differences.

This error cancellation allows the derived electron transfer rates ($2.62 \times 10^8 s^{-1}$) to achieve consistency with classical predictions ($1.89 - 2.49 \times 10^8 s^{-1}$), demonstrating that the approach remains viable for biological applications. The observed error cancellation in energy differences thus provides a strategic pathway for NISQ devices to contribute meaningfully to computational biochemistry by focusing on difference-based observables rather than absolute electronic energies.

3 Conclusion

In this work, we successfully developed and validated VQE-PDFT, a novel quantum-classical hybrid framework that integrates variational quantum eigensolver with multiconfiguration pair-density functional theory for strongly correlated electronic systems. Our approach employs quantum circuits for multiconfigurational wavefunction optimization while utilizing classical on-top density functionals for correlation energy calculation, achieving computational accuracy comparable to conventional MC-PDFT with significantly reduced quantum resource requirements.

Validation on the CT7/04 dataset demonstrated excellent agreement with reference calculations (MUE = 0.853 kcal/mol with respect to W-1 theory), confirming the method’s reliability for multiconfigurational systems. We also extended this framework to biological applications by developing specialized hardware-efficient ansatz circuits optimized for NISQ device constraints, achieving circuit depths of 4-6 compared to 45 for ROUCCSD while maintaining high accuracy.

Our comprehensive application to electron transfer in the European robin cryptochrome ErCRY4 protein yielded transfer rates ($0.944 \times 10^{10} s^{-1}$ for HEA

and $0.864 \times 10^{10} s^{-1}$ for ROUCCSD) that aligned well with experimental ultrafast spectroscopy measurements ($0.709 \times 10^{10} s^{-1}$), validating our quantum-classical approach for complex biological environments. Critically, we demonstrated the first successful calculation of biological electron transfer rates on real quantum hardware, obtaining reliable results ($2.62 \times 10^8 s^{-1}$) that compare favorably with noiseless simulations despite inherent quantum noise.

Moreover, this work directly mitigates three challenges for NISQ-era quantum biology applications: explicit treatment of multiconfigurational electronic correlations through purpose-designed shallow-depth quantum circuits, enhancing the ability of calculating strongly correlated system; systematic error cancellation through focus on energy differences rather than absolute energies, where we discovered that errors partially cancel in Marcus parameter calculations with quantum hardware, alleviating the quantum noise’s impact on derived observables; and integration within a scalable multiscale framework adaptable to future hardware improvements, minimizing software and algorithmic modification while maintaining utility.

Several limitations warrant future investigation. Firstly, current quantum hardware constraints may restrict active space sizes and computational accuracy. Secondly, efficient optimization and measurement grouping methods should be developed to lower the costs of quantum computation, enabling sampling across more protein conformations on superconducting quantum hardware, which was the main obstacle while conducting experimental validation. Thirdly, our separate treatment of indole rings on two tryptophan residues, while justified by weak electronic coupling, represents an approximation that could be improved by simultaneous evaluation of multiple rings to capture longer-range interactions and intermediate transfer states.

Future developments shall focus on constraining electronic populations during VQE optimization through penalty functions or Lagrangian multipliers with information from Mulliken population analysis, enabling treatment of multiple simultaneous transfer pathways. As quantum hardware develops, our scalable framework can accommodate increased qubit counts and improved fidelities for more comprehensive biological system calculations.

In conclusion, VQE-PDFT establishes a practical pathway for quantum computing applications in biochemistry, demonstrating that despite current limitations, quantum-classical hybrid approaches can provide reliable predictions for real biological systems and represent a significant milestone toward quantum utility in molecular sciences.

4 Methods

4.1 Quantum computing reduced density matrices

In VQE-PDFT, the VQE-based CASCI optimization is performed before the PDFT energy evaluation. Once the optimization is converged, the parameters of circuits would be freezed and passed for subsequently evaluating 1-RDM and 2-RDM. The matrix elements of 1-RDM and 2-RDM may be written as the expectation value in terms of creation/annihilation operators under the second quantization, such as for 1-RDM,

$$\gamma_{pq} = \langle \psi | a_p^\dagger a_q | \psi \rangle,$$

where the electronic state $|\psi\rangle$ could be efficiently reconstructed with the previously freezed circuit parameters. Therefore,

$$\gamma_{pq} = \langle \psi(\theta) | a_p^\dagger a_q | \psi(\theta) \rangle,$$

where

$$|\psi(\theta)\rangle = U(\theta) |0\rangle.$$

This enable us to compute each elements of 1-RDM by executing the circuit with freezed parameters and sampling the expectation value of the corresponding operators after fermion-to-qubit mapping such as the parity transformation.

The same procedure is also valid for computing 2-RDM.

4.2 Designing the empirical HEA

The hardware-efficient ansatz (HEA) circuits are designed based on the electronic structure characteristics of tryptophan indole systems, optimizing quantum resources while maintaining computational accuracy for NISQ devices.

Qubit reduction through symmetry constraints

The (4e,3o) closed-shell and (3e,3o) open-shell active spaces require 6 qubits under parity transformation. Exploiting α and β electron number conservation reduces this to 4 qubits, as qubits 3 and 6 (representing total α -electron and total electron occupation parity) remain fixed throughout calculations. This symmetry-based reduction halves quantum resource requirements while preserving all physically accessible states.

Circuit construction based on dominant states

Analysis of converged ROUCCSD wavefunctions reveals that only 9 computational basis states contribute significantly for both closed-shell and open-shell systems, with open-shell configurations requiring a more complex correlation. The circuits employ Ry and controlled-Ry for parameterized gates, with strategic qubit entanglement ensuring spin conservation. For example, in open-shell (3e,3o) systems with 2α and 1β electrons, the first two qubits must not access $|00\rangle$ (violating α -electron conservation) while qubits 3-4 must not access $|10\rangle$ (violating β -electron conservation), restricting the total accessible space to $3 \times 3 = 9$ states. The resulting CHEA (Figure 3A) and OHEA (Figure 3B) achieve depths of 4 and 6 layers respectively, compared to ROUCCSD’s 45-layer requirement.

This empirical approach represents a compromise necessitated by current NISQ hardware limitations rather than fundamental algorithmic constraints. The system-specific design optimized for tryptophan electronic structure patterns requires extensive reanalysis for different chemical environments or larger active spaces. As quantum hardware develops to support deeper circuits (hundreds of circuit-depths) with high fidelity, more general approaches like ROUCCSD may become preferable over these tailored solutions [45].

4.3 Computational details

Quantum hardware implementation

VQE optimizations were performed on classical simulators to avoid noise accumulation during iterative parameter updates. Once parameter convergence was achieved, the optimized parameters were saved and transferred to quantum hardware for reduced density matrix evaluations.

Hardware specification. Quantum computation employed a customized 13-qubit superconducting quantum device with the following performance metrics: single-qubit gate fidelity of 99.93%, two-qubit CZ gate fidelity of 99.13%, coherence times of $T_1 = 83.8 \mu s$ and $T_2 = 45.6 \mu s$, and readout fidelities of $F_0 = 98.35\%$ and $F_1 = 95.88\%$. The connecting topology is shown in Figure 7.

RDM matrix element evaluation. The computation of 1-RDM and 2-RDM matrix elements followed a systematic six-step protocol:

1. **Pauli operators preparation.** Measurement Pauli operators were obtained through parity transformation of the corresponding matrix elements.

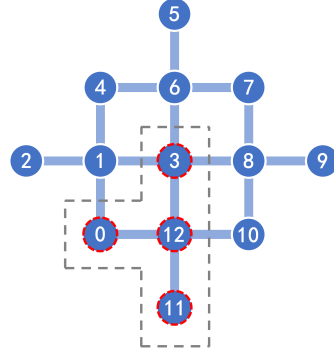


Fig. 7: Topology of 13-qubit superconducting quantum processor. During our validations, we only used 4 qubits indexing $\{0, 3, 11, 12\}$ and their couplings.

2. **Measurement grouping.** Compatible measurement operators were grouped using a simple grouping strategy to reduce quantum measurement overhead and noise impact. For instance, operators $Z_1 Z_2 I_3 I_4$ and $I_1 I_2 Z_3 Z_4$ can be simultaneously measured using the single operation $Z_1 Z_2 Z_3 Z_4$.
3. **Circuit compilation.** The empirical HEA circuits and grouped measurements were uploaded to the quantum hardware platform. These circuits then underwent qubit mapping and gate decomposition to match the chip topology and native gate set. As shown in Figure 7, the resulting compiled circuits utilized only 4 physical qubits, benefiting from our restrained use of two-qubit gates that eliminated the need for additional connectivity qubits.
4. **Circuit execution.** Each compiled circuit was executed with 2048 measurement shots, returning raw bit-string distribution data.
5. **Readout error mitigation.** Raw measurement data underwent readout error correction using the correlated Markovian noise model approach. This correction required calibration circuits executed once per single-point energy calculation with 8192 shots. The resulting calibration data was then stored for all subsequent measurements within that calculation. Finally, the error mitigation module processed raw bit-string distribution, calibration data, and target measurement operators to yield the corrected expectation value as the matrix elements.
6. **RDMs assembly.** Steps 1-5 were repeated for all required matrix elements to construct the entire 1-RDM and 2-RDM matrices.

This protocol enabled reliable extraction of reduced density matrices from noisy quantum hardware while maintaining computational efficiency through strategic measurement grouping and error mitigation.

Molecular dynamics sampling and statistical analysis

Twenty protein conformations were randomly sampled from molecular dynamics simulations of ErCRY4 [5]. Every indices of these conformations are detailed in Supplementary Table S2. Marcus theory parameters (reorganization energy λ , driving force ΔG^o , and electronic coupling $\langle |H_{DA}| \rangle$) were calculated independently for each conformation, with statistical averages of these parameters input into the Marcus rate expression. This approach ensures thermodynamically consistent averaging of Marcus theory parameters, representing the actual biological environment where conformational fluctuations modulate the underlying electronic structure properties.

Software implementation

The VQE-PDFT framework was implemented in Python through local modifications of PySCF [46] and TenCirChem [47] libraries. QM/MM calculations employed the ASH [48] framework for the link-atom and electrostatic embedding between the QM regions and the MM regions. Quantum hardware validation process was performed via the TensorCircuit [49] library for measurement grouping, communicating quantum device, and the readout error mitigation.

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Supplementary Information

Dimers	Dimer AS	m1 AS	m2 AS
$NH_3 - ClF$	(10e,10o)	(8e,8o)	(2e,2o)
$NH_3 - Cl_2$	(10e,10o)	(8e,8o)	(2e,2o)
$NH_3 - F_2$	(10e,10o)	(8e,8o)	(2e,2o)
$HCN - ClF$	(8e,8o)	(6e,6o)	(2e,2o)
$H_2O - ClF$	(10e,10o)	(8e,8o)	(2e,2o)
$C_2H_2 - ClF$	(10e,10o)	(10e,10o)	(2e,2o)
$C_2H_4 - F_2$	(10e,10o)	(10e,10o)	(2e,2o)

Table S1: Active space setting for calculating dissociation energies of Charge-Transfer dataset. The AS stands for Active Space. The m1 and m2 refers to the two monomers in the m1-m2 dimer, respectively.

frame	ΔG°	λ
157	0.099934408	0.549241928
264	0.043676293	0.347217505
461	0.040885297	0.220829577
631	0.163803428	0.517004701
682	0.038024409	0.495626829
706	0.071676358	0.307290925
777	0.037682775	0.547830361
1077	0.033809862	0.53029316
1107	0.167015044	0.532070337
1237	0.084752646	0.131341481
1243	0.023647298	0.623133927
1311	0.025645922	0.415820148
1368	0.05873396	0.774978721
1502	0.151100112	0.464021283
1603	0.113313698	0.217739833
1626	0.000816062	0.288449438
1627	0.021599893	0.345962916
1965	0.115245417	0.488976157
2071	0.13571487	0.340026962
2180	0.020222078	0.57379004

Table S2: ΔG° and λ calculations by empirical HEA on noiseless simulators. The 20 protein conformations were randomly sampled from the MD simulation of GS ErCRY4 protein in reference [5]. The frame numbers indicate locations of conformations during the entire simulation process. All results are given in eV.

frame	ΔG°	λ
157	0.075376004	0.533793409
264	0.066947358	0.550103409
461	0.022934202	0.529776474
631	0.098816411	0.654202407
682	0.102475986	0.552455689
706	0.109047811	0.593777179
777	0.091658923	0.600336874
1077	0.081086545	0.581411027
1107	0.082290769	0.627811309
1237	0.039260208	0.569951056
1243	0.110907047	0.655571383
1311	0.004307911	0.521179077
1368	0.035574098	0.544496837
1502	0.131879075	0.603524833
1603	0.003378448	0.516846946
1626	0.068215042	0.512984809
1627	0.036969471	0.529376022
1965	0.123957956	0.655946507
2071	0.043474286	0.524336485
2180	0.049975887	0.543812624

Table S3: ΔG° and λ calculations by ROUCCSD on noiseless simulators. The 20 protein conformations were randomly sampled from the MD simulation of GS ErCRY4 protein in reference [5]. The frame numbers indicate locations of conformations during the entire simulation process. All results are given in eV.

frame	H_{DA}	$ H_{DA} ^2$
157	-9.44011E-04	8.91158E-07
264	1.50656E-02	2.26972E-04
461	-3.43295E-02	1.17851E-03
631	-2.49168E-04	6.20846E-08
682	-2.21206E-03	4.89319E-06
706	-1.65424E-04	2.73650E-08
777	-4.04790E-04	1.63855E-07
1077	1.48556E-02	2.20688E-04
1107	-3.80092E-04	1.44470E-07
1237	1.37738E-03	1.89717E-06
1243	1.67444E-03	2.80376E-06
1311	5.25197E-03	2.75831E-05
1368	-1.37476E-03	1.88996E-06
1502	-1.84872E-02	3.41775E-04
1603	3.46027E-03	1.19735E-05
1626	2.59327E-03	6.72507E-06
1627	3.57321E-03	1.27678E-05
1965	-1.51863E-02	2.30625E-04
2071	-2.02746E-03	4.11059E-06
2180	3.41750E-03	1.16793E-05

Table S4: Electronic coupling calculations. The 6-31G(d) basis set and the CAM-B3LYP functional were used in all calculations. The 20 conformations were randomly sampled from the MD simulation of GS ErCRY4 protein in reference [5]. The frame numbers indicate locations of conformations during the entire simulation process. All results are given in eV.

index	system	shell type	Empirical HEA noiseless	ROUCCSD
1	point-a TrpB ⁺	open-shell	-362.99415	-362.99942
2	point-a TrpC ⁰	closed-shell	-363.26459	-363.26554
3	point-b TrpB ⁰	closed-shell	-363.26777	-363.26682
4	point-b TrpC ⁺	open-shell	-362.98880	-362.98758
5	point-c TrpB ⁰	closed-shell	-363.26994	-363.26920
6	point-c TrpC ⁺	open-shell	-362.99378	-362.99736
7	point-d TrpB ⁺	open-shell	-362.99775	-362.99268
8	point-d TrpC ⁰	closed-shell	-363.26632	-363.26518
λ (eV)			0.34003	0.52434
ΔG^o (eV)			0.13571	0.04347
k_{ET} ($10^8 s^{-1}$)			1.89064	2.48679

Table S5: Single-point energies, Marcus parameters, and the transfer rate of the random conformation. The conformation was randomly selected as the 2071 frame from the 20 conformations above and calculated within the QM/MM framework. The calculations were conducted by ROUCCSD and empirical HEA on noiseless simulator. The single-point energies are given in Hartree. The 6-31G basis set and the tPBE functional were used in all calculations.